25

Human evolution

Modern humans evolved from a *hominoid (hominid)*, or human-like, species (*L. hominus* human, *oid* in the form of). From fossil evidence, scientists have come up with an order of appearance of man's ancestors. The first man-like ape was *Ramapithicus*. Remains of these were found in Miocene beds in India and Africa. They were thought to date from around 14 million years ago. The remains had reduced canines and molars and slender incisors. (Note: most sources say that it was not a hominid.) *Ramapithicus* have two main differences to apes – stance and dentition (teeth). There has been modification to the pelvis and to feet to give an upright standing position, and apes have a rectangular jaw with large canines, whereas hominids have an arched and curved jaw with reduced canines.

1 Describe the function of canine teeth.

It is generally believed that humans originated in Africa from an ancestral line (existing 6 – 4 million years ago) known as *Australopiths*. Features were a cranial (brain) size of 450 mL, small canines and incisors, very large molars and a curved jaw. Also present were large eyebrow ridges, a sloping forehead but no definite chin. It is from *Australopiths* that the hominoids later evolved. *Australopiths* were found in two forms. The gracile (more delicate and graceful) species were *Australopithecus afarensis* and *A. africanus*. A stout and more muscular group, *A. robustus* and *A. boisei*, also existed. The exact relationship between the two forms (in evolutionary terms) is still being researched. Physical anthropologists (scientists who use physical features to piece together history) continue to work toward understanding human evolution.

The earliest hominoid, an omnivore dating about 2 million years ago, was *Homo habilis*. Surviving artefacts suggest that *H. habilis* was the first human-like species to use tools and modify their environment (eg creating shelters). Not all artefacts survived, in particular those made of biodegradable materials, so it is difficult to know exactly what other tools were available. One thing is clear, the fact that rocks from distant regions are found in areas where they are uncommon, suggests that this species travelled to collect the materials they wanted for tools. Facial features were similar to *Australopithecus*. Cranial capacity was 750 mL, with molars smaller and narrower than *Australopithecus*.

2 What is *cranial capacity*?

3 Define *artefact* and give an example.

4 Define *biodegradable* and suggest how a biodegradable material might be made into a tool.

5 Suggest ways in which tools and modifications to their surroundings might have made life easier for *H. habilis*.

6 How might these modifications have impacted on the environment? Explain your answer.

7 Outline the function of *molars*.

8 Define *omnivore*.

Homo erectus (*erect* straight, raised) is a descendant of *H. habilis*. *H. erectus* lived about 1.5 million to 300 000 years ago. Of all our distant ancestors, *H. habilis*, with its larger braincase, appears to be the most similar in skeletal structure to the modern human. Overall the species was much larger than people are today. It had big bones, teeth and a much larger forehead, but the resemblance is there.

H. erectus was a hunter-gatherer and is thought to have been the first to use fire. Fire may have initially been used to repel predators and then used for cooking and land management. The numbers and arrangement of fossils found has led scientists to believe that *H. erectus* was a social species that lived and died together. *Homo erectus* (sometimes called *Sinanthropus* or *Pithecanthropus*) existed about one million years ago. Fossils have been found in Java, China, Africa and Europe. Features include a low forehead, big eyebrow ridges, smaller molars and large incisors. Brain capacity was around 1000 mL.

Much of human evolution occurred during the Pleistocene era (also called the ice age) possibly due to the drastic changes in climate that took place at the time. About 300 000 to 200 000 years ago *Homo sapiens* arrived on the scene (*sapiens* wisdom, to know). Not all groups survived and debate continues as to when speech evolved.

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Homo neanderthalensis, or *Neanderthal* man, and *Cro-Magnon* man, are early human forms whose disappearances are still being questioned. Neanderthal man, named after the Neanderthal Valley Region (Rhine Province, Germany), existed about 120 000–35 000 years ago. Neanderthal man's range extended beyond Europe to Asia and Africa.

Evidence suggests that Cro-Magnon man may have been the same species as *H. sapiens*. Cro-Magnon man, dating to about 35 000 years ago, is best known for the artistic works they left on the walls of the Cro-Magnon caves (Dordogne, France). Modern humans date back to about 40 000 years ago. The type of evolution that has occurred to *H. sapiens* since this time is largely based on location and culture.

Archaic Homo sapiens existed in Europe (Neanderthal man), Africa and Southeast Asia around 35 000 to 120 000 years ago. With reduced molars, a slightly protruding jaw, pronounced eyebrows and a cranial capacity of 1400 mL – in most other respects was very similar to modern man. *Archaic Homo sapiens* were hunter-gatherers, had dwellings, buried the dead, used complex tools and fire.

Modern Homo sapiens first made an appearance about 40 000 years ago, for example Cro-Magnon man in Europe. Other fossil finds have been in North Africa, Southwest Asia and Australia. Cranial capacity was 1400 mL, a well-developed chin, high forehead and no eyebrow ridges were present. *Modern Homo sapiens* had a sophisticated culture, used complicated tools, lived in a variety of dwellings, used fire and practised burial of the dead.

9 Draw up a timeline with dates showing how our human ancestors appeared on Earth.

10 What has happened to brain size as we progress from ancient to more modern ancestors?

11 Why do we seem to know more about the later ancestors of humans than the earliest ones?

12 What are the main differences between hominids and apes?

13 Where did Neanderthal man exist?

14 Give one similarity and one difference between modern *Homo sapiens* and *Homo habilis*.

15 Are humans likely to evolve in the future? Explain your answer.

The development of human culture

All cultures have some kind of technology, social groups, arts and a spoken language.

> **Anthropology** *is the study of human beings, including their physical characteristics, culture, origins and distribution.*

16 There is much evidence to show how human culture has developed. Study the list below and arrange in chronological order, putting the earliest example first.

a Use of bronze.

b Carved bone items.

c Crops began to be grown.

d Domestication of cattle.

e Use of fire.

f Industrial revolution.

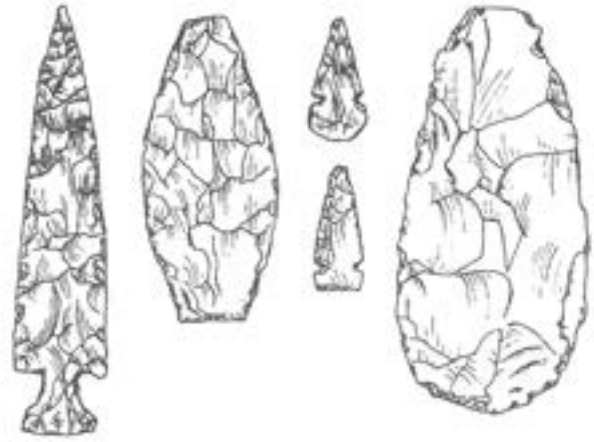
g Use of iron.

h Paintings – often of animals.

i Skins and wood in dwellings.

j Stone tools.

k Writing (on clay tablets).



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The development of culture and the physical changes in hominids over time is closely related. Brain, hand and artefacts (something created by humans for a practical purpose) developed together. Language is essential to transmit technical and social skills.

Many scientists spend a lot of time comparing the lives of modern humans and early hominids. There have been many changes since humans first walked our planet, but modern humans have much in common with our ancestors.

17 Outline three points of similarity with ancient humans and three points of difference.

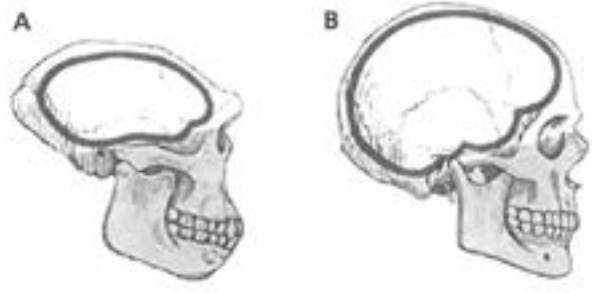
Main points in human evolution

Complete the following passage using these words:

agriculture, brain, chin, head, physical, posture, reduction, size, speech, tools

There are many **18** _____ changes associated with the evolution of humans. There has been an increased skull (braincase) **19** _____ and a steady increase in **20** _____ size (resulting in some reorganisation of brain structure). There has been an overall lengthening of the **21** _____ from a broad, rounded shape to a longer, oval shape and a **22** _____ in the size of the large ridges in the forehead (above the eyes). The jawbone has shortened and there has been an increase and lengthening of the **23** _____ area (previously the chin as we know it was minimal). Humans have also developed an upright **24** _____ with bipedal (walking on two legs) movement. It is not exactly known when **25** _____ was developed, but this was essential for communication. The making of **26** _____ led to nomadic hunters and hunter-gatherers. The introduction of **27** _____ increased food availability and allowed the formation of villages and towns.

28 Correctly label the two skulls shown here as *Homo sapiens* and *Homo erectus*.



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